Reading the Radar

Reference Document

Sebastián Torres
CIWRO, University of Oklahoma

Clouds4Africa Practicum
July 2026

Abstract

This document collects every symbol, formula, and concept used across the five "Reading the Radar" notebooks (Blocks 1–5). It is organized by block, in the same order the ideas are introduced in the practicum, and closes with an alphabetical glossary of field-specific terms, a master notation table, and a table of physical constants and reference relations. Use it as a standing reference while working through the widgets, or as a review sheet afterward.

Contents

1 Block 1 — The Beam in Space **3**
1.1 Notation . . . 3
1.2 Formulas . . . 3
1.3 Concepts . . . 5

2 Block 2 — One Measurement **5**
2.1 Notation . . . 5
2.2 Formulas . . . 6
2.3 Concepts . . . 7

3 Block 3 — Range, Velocity, and Ambiguity **8**
3.1 Notation . . . 8
3.2 Formulas . . . 8
3.3 Concepts . . . 9

4 Block 4 — Wavelength, Weather, and the Scan **10**
4.1 Notation . . . 10
4.2 Formulas . . . 10
4.3 Concepts . . . 11

5 Block 5 — Two Polarizations **12**
5.1 Notation . . . 12
5.2 Formulas . . . 12
5.3 Concepts . . . 14

6 Field-Specific Terms and Jargon **15**

7 Master Notation Table **19**

8 Physical Constants and Reference Values **20**

1 Block 1 — The Beam in Space

Block 1 concerns *where* the radar's beam actually goes: how wide it is, how high it climbs, how it can bend, and how terrain can hide part of it.

1.1 Notation

Symbol	Meaning	Notes
c	speed of light	2.998×10^8 m/s
a_e	true Earth radius	6371 km
λ	radar wavelength	$\lambda = c/f$
f	transmit frequency	GHz in these widgets
D	antenna (dish) diameter	metres
θ	half-power beamwidth	degrees; angular full-width at half the peak power
ϕ, elev	elevation (tilt) angle	degrees above horizontal
r	slant range	km, distance from radar along the beam
$h(r)$	beam-center height above the radar	km
k	effective Earth radius factor	$k = R_e/a_e$; $k = 4/3$ is "standard"
R_e	effective Earth radius	km; used to fold refraction into a straight-beam geometry
N	atmospheric refractivity	dimensionless (N-units), how much air slows radio waves
dN/dh	refractivity gradient	N-units/km; the quantity that actually sets k
M	modified refractivity	$M = N + 0.157\,h$ (h in metres); ducting $\iff dM/dh < 0$
PBB	partial beam blockage	fraction of the beam's power blocked at one gate
CBB	cumulative beam blockage	running-maximum of PBB out to the current gate
G	antenna gain	dimensionless (or dB); $G \propto (D/\lambda)^2$

1.2 Formulas

Half-power beamwidth (from dish size and wavelength):

$$\theta \approx 1.27\,\frac{\lambda}{D} \quad \text{(radians; convert to degrees as needed)} \tag{1}$$

A fixed dish gives a beamwidth fixed in *angle*, so the beam's physical cross-range width $w = r\theta$ *grows linearly with range.*

Antenna gain vs. beamwidth trade:

$$G \propto \left(\frac{D}{\lambda}\right)^2 \propto \frac{1}{\theta^2} \tag{2}$$

Halving the beamwidth (doubling D at fixed λ) raises gain by $20\log_{10}(2) \approx 6.02$ dB.

4/3-Earth beam-center height (standard refraction):

$$h(r,\phi) = \sqrt{r^2 + R_e^2 + 2rR_e\sin\phi} \; - \; R_e, \qquad R_e = \tfrac{4}{3}a_e \tag{3}$$

Effective Earth radius from the refractivity gradient:

$$R_e = \frac{a_e}{1 + a_e\,(dN/dh) \times 10^{-6}}, \qquad k = \frac{R_e}{a_e} \tag{4}$$

Standard atmosphere: $dN/dh \approx -39$ N-units/km $\Rightarrow k = 4/3$.

Paraxial (small-angle) beam height under any refraction gradient:

$$h_{\text{ap}}(r,\phi) \approx r\sin\phi + \frac{r^2}{2R_e} \tag{5}$$

This is the working formula in the anomalous-propagation widget; it agrees with the exact 4/3-Earth formula to well under 1% error out to elevations of 20° and ranges of a few hundred km.

Ducting criterion (using modified refractivity $M = N + 0.157h$, h in metres):

$$\text{ducting} \iff \frac{dM}{dh} < 0 \iff \frac{dN}{dh} < -157 \text{ N-units/km} \tag{6}$$

This is exactly the gradient at which $R_e \to \infty$ (denominator of Eq. 4 vanishes) and beyond which R_e turns negative — the beam curves as sharply as, or more sharply than, the Earth itself.

Ground-strike range under ducting (solve $h_{\text{ap}} = 0$ in Eq. 5 for $R_e < 0$):

$$r_{\text{strike}} = -2R_e\sin\phi \tag{7}$$

This is an *exact* solution of the paraxial model, not merely a small-angle approximation of it. Using a small-angle approximation $\sin\phi \approx \phi$, it's easy to see that doubling ϕ at fixed R_e doubles the strike range.

Partial and cumulative beam blockage (Bech et al. 2003, simplified disc-overlap form): with $y = h_{\text{terrain}} - h_{\text{beam}}$ and beam radius $a = r\theta/2$,

$$\text{PBB} = \frac{y\sqrt{a^2 - y^2} + a^2\arcsin(y/a) + \pi a^2/2}{\pi a^2}, \qquad \text{CBB}(r) = \max_{r' \le r} \text{PBB}(r') \tag{8}$$

PBB is the fraction of the beam's circular cross-section hidden by terrain at one gate; CBB is the worst blockage encountered anywhere along the ray up to that gate, and it never decreases with range even after the terrain drops away, because the blocked energy is gone for good.

The Z–R relation (used here for the first time; it returns in Blocks 4–5): an *empirical* power law connecting reflectivity to rain rate,

$$Z = 200\,R^{1.6} \qquad (Z \text{ in mm}^6\,\text{m}^{-3},\ R \text{ in mm/h}) \tag{9}$$

There is no unique physical mapping from Z to R — reflectivity depends on drop size to the sixth power while rain rate depends on it much more weakly, so different drop populations can share a Z yet rain at different rates (Block 5 resolves this with Z_{DR}). The coefficients $(200, 1.6)$ are the classic

Marshall–Palmer-type values for "average" stratiform rain, and are the ones used throughout these notebooks; operational systems switch coefficients by rain regime.

Blockage bias in reflectivity and rainfall: remaining fractional power is $(1 - \mathrm{CBB})$, so

$$\Delta Z_{\mathrm{dB}} = 10 \log_{10}(1 - \mathrm{CBB}) \tag{10}$$

and, propagating through the Z–R relation $Z = 200R^{1.6}$,

$$\frac{R_{\mathrm{observed}}}{R_{\mathrm{true}}} = (1 - \mathrm{CBB})^{1/1.6} \tag{11}$$

1.3 Concepts

Beam as a volume, not a ray
: The radar's illumination pattern has finite angular width and, therefore, a physical cross-section that grows with range; "no echo" along a ray never means "no weather," only "nothing detected along this finite-width, possibly-obstructed cone."

Beam climb
: Under standard refraction the beam's curvature is gentler than the Earth's, so even a 0° beam rises above the surface with range; low-level features far from the radar are systematically sampled from higher altitude than one might assume.

Anomalous propagation (AP)
: Any departure from the standard −39 N/km refractivity gradient. Sub-refraction bends the beam less (it overshoots); super-refraction and ducting bend it more (it stays low or curves into the ground).

Ducting / trapping
: The extreme end of super-refraction, where the beam bends as sharply as the Earth curves and becomes trapped near the surface, producing false "ground clutter" echoes with no associated weather. Typically caused by a nighttime temperature inversion, which is why ducting artifacts appear and vanish with the diurnal cycle.

Terrain blockage
: Partial or total obstruction of the beam by nearby high ground. A radial affected by blockage reads a biased-low reflectivity (and therefore underestimated rainfall) at every gate downrange of the obstruction, not just at the obstruction itself.

Volume Coverage Pattern context
: Blockage and AP are both reasons a single low tilt cannot be trusted everywhere; Block 4 shows how operators trade tilt height against these very problems.

2 Block 2 — One Measurement

Block 2 asks how big a single radar value actually is (the resolution volume) and whether the radar can detect a given echo at all (the radar equation).

2.1 Notation

Symbol	Meaning	Notes
τ	pulse length (duration)	microseconds
Δr	range resolution	$\Delta r = c\tau/2$; also the depth of the resolution volume
P_t	peak transmit power	kW
Z	radar reflectivity factor	dBZ; proportional to $\sum D^6$ over scatterers in the volume
k_B	Boltzmann's constant	1.380649×10^{-23} J/K
T_0	receiver noise reference temperature	290 K
NF	receiver noise figure	dB
N	receiver thermal noise power	W; $N = k_B T_0 \, \tau^{-1} \, 10^{NF/10}$. (Not Block 1's refractivity N — an unfortunate but conventional symbol collision; context always disambiguates.)
η	antenna efficiency	dimensionless, ~ 0.45 typical
$\lvert K_w \rvert^2$	dielectric factor of water	≈ 0.93 at radar wavelengths
$\mathrm{SNR}_{\min}$	minimum detectable signal-to-noise ratio	dB, e.g. ~ 3 dB
$Z_{\min}(r)$	minimum detectable reflectivity	dBZ; the sensitivity floor as a function of range
V	resolution volume	km^3; $\approx (r\theta)^2 \times \Delta r$

2.2 Formulas

Range resolution / sample depth from pulse length:

$$\Delta r = \frac{c\tau}{2} \tag{12}$$

Two targets closer together than Δr blur into a single echo.

Blind range: the radar cannot listen while still transmitting, so it is blind out to approximately Δr as well — a longer pulse (more sensitive) also means a larger dead zone close to the radar.

Sensitivity scaling with pulse length: received energy per pulse $\propto \tau$, so

$$\Delta(\text{sensitivity})_{\text{dB}} = 10 \log_{10}\left(\frac{\tau_2}{\tau_1}\right) \tag{13}$$

Doubling τ buys +3.01 dB of sensitivity but doubles Δr — resolution and sensitivity move in opposite directions from the same knob.

Resolution volume (beam cross-section from Block 1 combined with pulse depth):

$$V(r) \;\approx\; \underbrace{(r\theta)^2}_{\text{cross-beam area}} \times \underbrace{\frac{c\tau}{2}}_{\text{depth}} \;\propto\; r^2 \tag{14}$$

The cross-beam dimensions grow with range while the depth stays fixed by the pulse, so the whole sampled volume grows as r^2 — a distant pixel is a much larger spatial average than a near one.

Weather radar equation (distributed-target form; received power $\propto Z/r^2$, so the minimum *detectable* Z climbs as r^2):

$$Z_{\min}(r) = 10\log_{10}\left(\frac{\mathrm{SNR}_{\min}\cdot N\cdot r^2}{K}\Big/10^{-18}\right)\ \mathrm{dBZ} \tag{15}$$

with thermal noise power spectral density $N = k_B T_0 \tau^{-1}\cdot 10^{NF/10}$ and the radar-constant term

$$K = \frac{\pi^3\, c\, P_t\, G^2\, \theta^2\, \tau\, |K_w|^2}{1024\,\ln 2\,\lambda^2}, \qquad G = \eta\left(\frac{\pi D}{\lambda}\right)^2, \quad \theta = 1.27\frac{\lambda}{D} \tag{16}$$

This is the algebraic combination of Blocks 1–2's ideas (gain from dish/wavelength, beamwidth, pulse length) into a single sensitivity curve. Because it derives from a *distributed* target filling the resolution volume, $Z_{\min} \propto r^2$ (+6 dB per range doubling) — gentler than the $1/r^4$ falloff of a discrete point target (bird, aircraft), whose echo does not grow to fill a bigger volume as range increases.

Wavelength dependence of sensitivity at fixed dish: since $G \propto \lambda^{-2}$ and $\theta \propto \lambda$, the λ^2 in K's numerator and denominator combine so that $Z_{\min} \propto \lambda^4$ at fixed D — a $1/\lambda^4$ intrinsic sensitivity advantage for shorter wavelengths, which is the reason a compact X-band radar can rival a much larger S-band dish for raw detectability (before attenuation, see Block 4).

2.3 Concepts

Resolution volume
: The chunk of atmosphere a single reported value actually averages over: fixed depth $c\tau/2$ set by the pulse, growing cross-beam width $r\theta$ set by the beam. Every reported number is a spatial average over this volume, not a point sample.

Blind range
: The zone immediately around the radar (extending out to roughly $\Delta r = c\tau/2$) where nothing can be observed, because the receiver cannot listen while the transmitter is still firing. It grows with pulse length, so the same knob that buys sensitivity also enlarges the near-radar dead zone — a shallow feature close to the radar can be unobservable on a long-pulse scan.

Beam filling
: When the resolution volume's cross-beam width exceeds the physical size of the feature being sampled (e.g. a compact convective core), the reported peak reflectivity is diluted by averaging in the weaker surroundings, biasing the observed peak low.

Minimum detectable signal (sensitivity)
: The weakest reflectivity a radar can distinguish from receiver noise at a given range; it degrades (gets less sensitive, i.e. $Z_{\min}$ rises) with range because received power falls as $1/r^2$ while the noise floor does not.

Detection vs. absence
: Because $Z_{\min}$ rises with range, a real but weak echo far from the radar can fall below the noise floor and simply not appear — "no echo" can mean "too faint to detect," distinct from Block 1's "no echo because blocked."

3 Block 3 — Range, Velocity, and Ambiguity

Block 3 is the time-domain twin of Block 1: the same pulse train sets both how far the radar can see unambiguously and how fast a target it can measure, and the two limits trade directly against each other.

3.1 Notation

Symbol	Meaning	Notes
PRF	pulse repetition frequency	kHz; how often a pulse is transmitted
PRT	pulse repetition time	ms; PRT = 1/PRF
$r_{\max}$	maximum unambiguous range	km
v_{Nyq}, $v_{\max}$	Nyquist (maximum unambiguous) velocity	m/s
v_{true}	true radial velocity of a scatterer	m/s (positive = away from radar)
v_{disp}	displayed (aliased) velocity	m/s
v_{ext}	extended Nyquist velocity from dual-PRF	m/s

3.2 Formulas

Maximum unambiguous range: a pulse must return before the next is transmitted, or its echo is mistaken for a much closer, later pulse's echo (a second-trip echo):

$$r_{\max} = \frac{c}{2\,\text{PRF}} \tag{17}$$

Range folding: a target truly at range $r_{\text{true}} > r_{\max}$ is displayed at

$$r_{\text{disp}} = r_{\text{true}} \bmod r_{\max} = r_{\text{true}} - r_{\max} \quad (\text{for } r_{\max} < r_{\text{true}} < 2r_{\max}) \tag{18}$$

Nyquist velocity:

$$v_{\text{Nyq}} = \frac{\lambda\,\text{PRF}}{4} \tag{19}$$

The Doppler dilemma: multiplying the two ceilings eliminates PRF, leaving a constant set purely by wavelength:

$$r_{\max} \cdot v_{\text{Nyq}} = \frac{c\lambda}{8} \tag{20}$$

A single PRF places the radar at one point on this fixed curve: raising PRF buys velocity range and spends unambiguous range, and vice versa. Because the product scales with λ, moving to a shorter wavelength (e.g. C-band vs. S-band) shrinks the achievable $r_{\max} \cdot v_{\text{Nyq}}$ envelope proportionally — the dilemma gets *worse*, not better, at C-band.

Velocity aliasing (folding): a true velocity outside $\pm v_{\text{Nyq}}$ is displayed folded back into that window,

$$v_{\text{disp}} = \left[(v_{\text{true}} + v_{\text{Nyq}}) \bmod 2v_{\text{Nyq}}\right] - v_{\text{Nyq}} \tag{21}$$

A storm faster than v_{Nyq} can therefore be displayed with the wrong speed, or even the wrong sign (apparently inbound when truly outbound).

Velocity dealiasing: recover the true velocity by adding whole multiples of $2v_{\text{Nyq}}$,

$$v_{\text{true}} = v_{\text{disp}} + 2n\, v_{\text{Nyq}}, \qquad n \in \mathbb{Z} \tag{22}$$

choosing n using continuity with neighboring, unambiguous gates. The method fails when an *entire* region exceeds the Nyquist velocity, because there is no nearby unambiguous gate left to anchor the continuity assumption.

Dual-PRF extended Nyquist velocity: transmitting (and combining) two PRFs with individual Nyquist velocities $v_1 > v_2$ extends the effective unambiguous velocity to

$$v_{\text{ext}} = \frac{v_1 v_2}{v_1 - v_2} \tag{23}$$

while retaining the *longer* of the two unambiguous ranges (that of the lower PRF) — this genuinely moves the radar off the single-PRF curve $r_{\max} v_{\text{Nyq}} = c\lambda/8$, buying both range and velocity coverage simultaneously (at the cost of added system complexity and its own dealiasing artifacts).

3.3 Concepts

Radial velocity
: The only component of motion a Doppler radar measures: the part of a target's velocity *along the beam*, toward or away from the radar. Motion *across* the beam produces no Doppler shift and is invisible. This is why a rotating storm shows the characteristic inbound/outbound couplet (flow toward the radar on one side, away on the other), separated by a near-zero band — the *zero isodop* — wherever the wind blows perpendicular to the beam, even though the actual wind speed there is far from zero.

Second-trip echo
: An echo from beyond $r_{\max}$ that returns after the next pulse has already gone out, and is consequently displayed at the wrong (shorter) range. Because the radar computes reflectivity using the (wrong) displayed range rather than the true range, and calibration compensates for the $1/r^2$ power falloff using the assumed range, a second-trip echo's reported Z is also biased (too high, since the true range is greater than the range used in the correction).

Range–velocity trade / Doppler dilemma
: The fundamental limit $r_{\max} v_{\text{Nyq}} = c\lambda/8$: a single PRF cannot deliver long unambiguous range and high unambiguous velocity at the same time.

Velocity aliasing
: The Doppler analogue of range folding: velocities beyond the Nyquist limit wrap around and can appear as an abrupt, unphysical reversal in an otherwise smooth velocity field — a diagnostic clue for telling aliasing from a genuine wind shear or shear line.

Dual-PRF
: An operational technique that interleaves two pulse rates to extend the usable Nyquist velocity well beyond what either PRF alone could give, without sacrificing as much unambiguous range.

4 Block 4 — Wavelength, Weather, and the Scan

Block 4 is the synthesis block: it revisits every prior trade-off through the lens of *choice*. Operators and designers pick a wavelength and a scan strategy, and physics charges a coupled price for each choice.

4.1 Notation

Symbol	Meaning	Notes
a, b	attenuation coefficients	in $k = aR^b$ (dB/km, one-way); band-dependent
$k(R)$	one-way specific attenuation	dB/km, as a function of rain rate R
PIA	path-integrated attenuation	dB; two-way loss accumulated along the beam
VCP	volume coverage pattern	a defined stack of elevation tilts scanned in sequence
rpm	antenna rotation rate	revolutions per minute
T_{vol}	volume update time	seconds; time to complete one full VCP

4.2 Formulas

Path-integrated (two-way) attenuation through a rain cell along the beam:

$$\text{PIA}(x) = 2 \int_0^x a\, R(x')^b \, dx' \tag{24}$$

the factor of 2 accounts for the wave crossing the rain on the way out *and* on the way back. Because it is a path integral, the loss behind a cell depends on everything the beam has already crossed, and accumulates over the entire path — a strongly attenuating cell can render anything behind it, including a second storm, effectively invisible. The exponent b matters as much as the coefficient a: the widgets use $b = 1$ at S- and C-band but $b = 1.3$ at X-band, so X-band's loss grows *super-linearly* with rain rate — doubling a core's intensity more than doubles the shadow behind it, which is precisely why intense convection punishes X-band disproportionately. (The values used throughout are collected in the reference table at the end of this document.)

Observed vs. true reflectivity:

$$Z_{\text{observed}}(x) = Z_{\text{true}}(x) - \text{PIA}(x) \tag{25}$$

Intrinsic sensitivity vs. wavelength (fixed dish): from Block 2's radar equation, $Z_{\min} \propto \lambda^4$ at fixed D, so a shorter wavelength is intrinsically more sensitive by a fixed number of dB at *every* range. The attenuation penalty, by contrast, is a path integral and *grows with range* — so a fixed head-start (sensitivity) is eventually overtaken by a penalty that keeps accumulating (attenuation), and for sufficiently distant or heavy rain the longer wavelength wins regardless of the shorter wavelength's initial advantage.

Effective detectability curve combining both effects:

$$Z_{\text{eff, min}}(r) = \underbrace{Z_{\text{min}}(r)}_{\text{intrinsic, Block 2}} + \underbrace{2\,a\,R^b\,r}_{\text{two-way attenuation to range } r} \tag{26}$$

Volume update time: with N_{tilts} elevations in a VCP and rotation rate rpm (assuming one full 360° sweep per tilt),

$$T_{\text{vol}} = N_{\text{tilts}} \times \frac{60}{\text{rpm}} \ \text{seconds} \tag{27}$$

Cone of silence and low-level gap: the highest tilt in a VCP leaves everything above its beam unsampled (the cone of silence, widening with height above the radar); the lowest tilt leaves a low-level gap below its beam that also widens with range (Block 1's beam-height formula). Adding tilts narrows both gaps but lengthens T_{vol} — a strict three-way trade among vertical coverage, data quality (achievable via more pulses per tilt), and update speed.

Power–gain–wavelength entanglement: received power for a distributed target scales as

$$P_r \propto \frac{P_t}{\theta^2 \lambda^2} \tag{28}$$

so two radars that differ in transmit power, beamwidth, *and* wavelength simultaneously cannot have their sensitivity advantage cleanly attributed to any one factor from a side-by-side image alone; only a controlled comparison at fixed dish size (as in the detectability widget) isolates the $1/\lambda^4$ wavelength effect on its own, since at fixed D, $\theta \propto \lambda$ folds one extra λ^2 into the denominator.

4.3 Concepts

Attenuation
: Loss of beam energy to absorption and out-of-beam scattering by precipitation along the path. Negligible at S-band, operationally significant at C-band, and potentially severe (capable of extinguishing the beam entirely) at X-band.

Sensitivity–attenuation competition
: Shorter wavelengths are simultaneously more sensitive (helps detect faint, distant echoes) and more attenuated (hurts detecting anything *behind* heavy rain); which band "wins" for a given faint, distant target depends on how much rain lies in between.

Volume Coverage Pattern (VCP)
: A predefined, named set of elevation tilts scanned in sequence to build one 3-D radar volume; different VCPs trade vertical resolution/coverage against update speed (e.g. a fast 3-tilt low-level VCP vs. a slow, dense clear-air VCP). The numbered patterns in the widget (VCP 12, VCP 21, VCP 31) are NEXRAD's actual operational designations.

Cone of silence
: The unsampled, cone-shaped region directly above the radar, above the highest tilt in the VCP; a tall storm passing close to the radar can have its upper portion (echo tops, overshooting tops) entirely inside it. Widens with height, and shrinks only by adding higher tilts — at the cost of volume time.

Low-level gap
: The counterpart below the *lowest* tilt: because even a 0.5° beam climbs with range (Block

1), the layer between the surface and the lowest beam grows with distance, and shallow, low-level weather far from the radar goes unsampled no matter how many upper tilts are added.

Coupled cost
The block's unifying idea: every design or operational choice — wavelength, dish size, pulse length, PRF, number of tilts — buys an improvement in one dimension only by spending capability in another; there is no free upgrade.

5 Block 5 — Two Polarizations

Blocks 1–4 all use a single (horizontal) polarization, which tells you *how much* echo there is. Block 5 adds a second, vertically-polarized channel, and it is the *comparison* between the two that reveals what the scatterers actually are.

5.1 Notation

Symbol	Meaning	Notes
H, V	horizontal / vertical polarization channels	the two transmitted/received wave states
Z_{DR}, ZDR	differential reflectivity	dB; $10\log_{10}(Z_H/Z_V)$ in linear power, reported in dB
ρ_{hv}	co-polar correlation coefficient	dimensionless, 0–1; how alike the H and V returns are across the sample
Φ_{DP}	differential phase	degrees; accumulated H–V phase difference along the path
K_{DP}	specific differential phase	deg/km; $K_{DP} = \frac{1}{2}\, d\Phi_{DP}/dr$, the range-derivative (slope) of Φ_{DP}
D	drop diameter	mm
N	drop number concentration	the widget's "log N (amount)" slider; how *many* drops are in the volume. (A third distinct use of the symbol N; see Blocks 1–2.)
b/a	raindrop axis ratio (vertical/horizontal)	dimensionless; $\to 1$ for small drops, < 1 (flattened) for large ones
$\varepsilon - 1$	water dielectric factor (real part)	sets the magnitude of the ZDR–size relationship
R	rain rate	mm/h

5.2 Formulas

Differential reflectivity (H vs. V power ratio, in dB):

$$Z_{DR} = 10\log_{10}\left(\frac{Z_H}{Z_V}\right) \tag{29}$$

For an oblate (flattened) raindrop with depolarization factors L_x (horizontal) and L_z (vertical) derived from its axis ratio, and $k = 1/(\varepsilon - 1)$,

$$Z_{DR} = 20 \log_{10}\left(\frac{L_z + k}{L_x + k}\right). \tag{30}$$

This is the Rayleigh–Gans oblate-spheroid result (Seliga and Bringi 1976). Z_{DR} is a *ratio* of H to V power: it depends only on drop shape/size, not on how many drops are present, so it is unaffected by concentration or calibration errors that scale H and V equally — unlike Z, which depends on both size *and* number.

Empirical raindrop axis ratio used in the widget:

$$\frac{b}{a}(D) \approx 1.03 - 0.062\, D \quad (D \text{ in mm, clipped to } [0.45, 1.0]) \tag{31}$$

Reflectivity from a drop population, and the size–number degeneracy: for drops of a single size D at concentration N,

$$Z \;\propto\; N\, D^6 \tag{32}$$

so raising the *amount* of rain by a factor of 10 ($\log N$ up by 1) lifts Z by exactly +10 dB while leaving Z_{DR} untouched. Because Z mixes size and number in one product, it cannot distinguish many small drops from few large ones — but Z_{DR} depends on D alone, so measuring both pins down D first (from Z_{DR}) and then N (from Z): a two-equation, two-unknown retrieval of the drop-size distribution that single-polarization Z can never perform.

Correlation coefficient: for H and V per-particle scattering amplitudes s_h, s_v across the resolution volume,

$$\rho_{hv} = \frac{|\langle s_h s_v^* \rangle|}{\sqrt{\langle |s_h|^2 \rangle \langle |s_v|^2 \rangle}} \tag{33}$$

$\rho_{hv} \approx 1$ when every scatterer in the volume is the same type (e.g. all rain); mixing particle types (hail among rain, melting snow, or non-weather returns such as birds, insects, chaff, or ground clutter) lowers it. Beam broadening with range (Block 2) also mixes more varied targets into one volume and can lower ρ_{hv} on its own, so ρ_{hv} must always be interpreted alongside the sample (resolution) volume size.

Differential phase and its slope: Φ_{DP} accumulates monotonically along the beam path through anisotropic (non-spherical) scatterers, and never decreases; its along-range slope defines specific differential phase,

$$K_{DP} = \frac{1}{2}\frac{d\Phi_{DP}}{dr} \quad (\text{deg/km}) \tag{34}$$

Because Φ_{DP} is a *phase* measurement rather than a power measurement, it is unaffected by attenuation, calibration drift, or partial beam blockage — all of which corrupt Z.

Rain rate from KDP (empirical power law used in the widget, exactly invertible):

$$R = 44\, K_{DP}^{0.82} \qquad \Longleftrightarrow \qquad K_{DP} = \left(\frac{R}{44}\right)^{1/0.82} \tag{35}$$

K_{DP}-based rain-rate estimates are noisy in light rain (little phase accumulates over a short path, so the slope is poorly determined) but excel in heavy rain — precisely where Z-based estimates are

most compromised by attenuation (Block 4). Operationally, algorithms blend Z, Z_{DR}, and K_{DP} to cover the full range of rain intensities.

Differential attenuation: because raindrops are flattened, the horizontally polarized wave is attenuated slightly *more* than the vertical one crossing heavy rain — so behind a strong cell, not only is Z biased low (Block 4's shadow), but Z_{DR} is biased low as well, since the H and V channels are no longer weakened equally and even their *ratio* is corrupted. This is why, behind heavy rain, both power-based variables (Z and Z_{DR}) fail together while the phase-based pair (Φ_{DP}, K_{DP}) sails through untouched — the deepest reason K_{DP} is the rainfall workhorse on attenuating (C- and X-band) networks.

5.3 Concepts

Dual-polarization radar
: A radar that transmits and receives both horizontal and vertical polarizations, enabling three additional variables (Z_{DR}, ρ_{hv}, Φ_{DP}/K_{DP}) beyond single-pol reflectivity Z.

Differential reflectivity (Z_{DR})
: Encodes the *shape and size* of scatterers, via the differential flattening of large raindrops; near-zero or negative for round targets (small drops, tumbling hail), positive and growing for large, flattened raindrops.

Correlation coefficient (ρ_{hv})
: The radar's "purity meter": high and uniform for homogeneous rain, reduced by any mixture of particle types — making it the most direct single variable for spotting non-meteorological echoes and the melting layer (bright band).

Specific differential phase (K_{DP})
: A phase-based rain-rate estimator immune to attenuation and calibration error; the dual-pol remedy for the very shadow that Block 4 shows single-pol reflectivity suffering behind heavy rain, especially at C-band.

Drop-size distribution (DSD)
: The population of raindrops in a volume, described (at minimum) by a characteristic size and a number concentration. Z alone collapses the two into one number ($Z \propto ND^6$); adding Z_{DR} separates them, which is the physical basis for dual-pol's improved rainfall estimation.

Hydrometeor classification
: The practice of combining Z, Z_{DR}, and ρ_{hv} (and often K_{DP}) into a signature that identifies what is producing an echo — e.g. high Z + moderate/high Z_{DR} + high ρ_{hv} points to heavy rain, not hail, even though Z alone cannot make that distinction.

6 Field-Specific Terms and Jargon

The formulas and notation above cover the quantitative machinery; the list below covers the surrounding vocabulary — named radars, scan modes, echo types, and shorthand — used throughout the notebooks but not necessarily spelled out where they first appear. Entries are alphabetical, with the block(s) where each term is used noted at the end.

Azimuth The horizontal pointing direction of the beam, measured as a compass angle; together with elevation angle and range, it is one of the three coordinates (Block 1) that locate every radar measurement in space. *(Blocks 1, 3, 5)*

Beam (radar beam)
The finite-width cone of electromagnetic energy the antenna transmits and receives along; treated throughout as a volume rather than an infinitely thin ray. *(Blocks 1–5)*

Bright band
See **Melting layer**.

Boundary layer
The lowest kilometre or two of the atmosphere, directly stirred by the surface (heating, friction, moisture); the home of the shallow, weak, clear-air echoes that long-pulse scans hunt — and of features that can sit inside a radar's blind range or below its lowest beam. *(Block 2)*

Clear-air mode
A radar operating mode (as opposed to precipitation mode) tuned to maximize sensitivity to weak, non-precipitating targets — insects, refractive-index turbulence, dust — typically using a longer pulse and accepting a larger blind range and coarser resolution in exchange for the extra sensitivity. *(Block 2)*

Clutter (ground clutter)
Radar returns from stationary, non-meteorological surface targets (terrain, buildings, towers) rather than from weather; a common false-echo source, especially under anomalous propagation/ducting, which bends the beam into terrain it would normally clear. *(Blocks 1, 5)*

Convective core
The most intense part of a thunderstorm or shower, typically compact (on the order of a few km across) and associated with the storm's heaviest rain, largest drops, or hail. *(Blocks 2, 4)*

dBZ The decibel unit in which radar reflectivity factor Z is conventionally reported: $\mathrm{dBZ} = 10\log_{10}(Z)$, where Z itself is in units of $\mathrm{mm}^6/\mathrm{m}^3$. Used throughout as the standard scale for reflectivity, from deep negative values (undetectable) through the tens (light rain) to 60+ (very heavy rain or hail). *(Blocks 1–5)*

Dish Colloquial shorthand for the parabolic reflector antenna that focuses the transmitted beam and collects the returned echo; its diameter D is one of the two quantities (with wavelength) that set beamwidth and gain. *(Blocks 1, 2, 4)*

Doppler radar
A radar that measures not only the strength of the returned echo (reflectivity) but

also the frequency shift imparted by a moving target, from which radial velocity is derived; the basis for all of Block 3's range–velocity material. *(Blocks 3, 5)*

Echo top The highest altitude at which a storm returns detectable echo — a common proxy for storm depth and intensity. Estimated echo tops inherit every sampling limitation in these notebooks: they are quantized by the tilt spacing, smeared by the growing resolution volume at long range, and capped by the cone of silence close to the radar. *(Blocks 2, 4)*

Elevation angle / tilt
The vertical pointing angle of the beam above the horizon; "tilt" is used interchangeably with elevation angle throughout, especially when referring to one specific angle within a multi-tilt scan strategy (VCP). *(Blocks 1, 4, 5)*

Gap-filler radar
A small, often mobile or low-cost radar (frequently X-band, with a correspondingly small dish) deployed to cover a gap in a larger network's coverage, such as a data-sparse region or an area shadowed by terrain from the primary radar. *(Block 1)*

Gate (range gate)
One discrete sample interval along the beam, set by how finely the received signal is digitized in time (and hence range); a full radial is built from many gates, and a full sweep from many radials, each at a different azimuth. *(Blocks 1–5)*

Hydrometeor
Any water-based particle in the atmosphere that a radar can detect — raindrops, snowflakes, ice crystals, hail — as distinct from non-meteorological scatterers such as birds, insects, or ground clutter. *(Block 5)*

Melting layer (bright band)
The altitude band where falling snow/ice melts into rain; it produces a locally enhanced single-pol reflectivity (the "bright band," from the abrupt jump in particle density and dielectric properties as ice becomes water-coated) and a locally reduced ρ_{hv} (from the mixture of ice, mixed-phase, and liquid particles), making dual-pol a cleaner detector of it than reflectivity alone. *(Blocks 1, 5)*

NEXRAD The U.S. Next-Generation Weather Radar network (technically the WSR-88D radars) — the S-band, 8.5 m-dish, 750 kW operational radar whose parameters are used as the recurring "default" configuration throughout these notebooks. Individual sites carry four-letter identifiers, several of which appear in the real-data widgets: KTLX (Oklahoma City), KINX (Tulsa), KNKX (San Diego), KAKQ (Wakefield, VA). *(Blocks 1, 2, 4)*

Non-meteorological echo
A radar return from something other than precipitation — birds, insects, ground clutter, chaff (radar-reflective military countermeasure material), or other debris — typically flagged by low ρ_{hv} (Block 5) because such targets are far less uniform, particle to particle, than natural rain. *(Blocks 1, 5)*

PPI (Plan Position Indicator)
The familiar circular radar display built by sweeping the beam through a full 360° in

azimuth at one fixed elevation angle; the basic image format referenced throughout when discussing what a radar "shows." *(Block 1)*

Precipitation mode
The counterpart to clear-air mode: a scanning/pulse configuration tuned for observing precipitation, typically using shorter pulses (finer resolution, less sensitivity to very weak targets) and, often, faster update rates. *(Block 2)*

QPE (Quantitative Precipitation Estimation)
The practice of converting radar-observed variables (Z, and in dual-pol also Z_{DR}/K_{DP}) into an estimated rain rate or accumulation; referenced whenever the notebooks discuss how blockage, attenuation, or beam height errors translate into rainfall error. *(Blocks 1, 4, 5)*

Radial All the gates along one beam direction — a single "ray" of the display at one azimuth. Effects tied to a fixed direction (terrain blockage, attenuation shadows) appear as *radial* features: the "spokes" of missing echo in Block 1's data case are blocked radials. *(Blocks 1, 4, 5)*

S-band / C-band / X-band
Conventional radar frequency-band names, in order of decreasing wavelength: S-band (∼2–4 GHz, $\lambda \approx 10$ cm, e.g. NEXRAD), C-band (∼4–8 GHz, $\lambda \approx 5$ cm, common on many national/regional networks including much of Africa and Europe), and X-band (∼8–12 GHz, $\lambda \approx 3$ cm, common on compact, mobile, or gap-filler radars). The practicum's recurring wavelength trade-offs (sensitivity vs. attenuation) are framed around these three bands. *(Blocks 1, 2, 4)*

Sweep / volume scan
A sweep is one full 360° azimuthal rotation at a fixed elevation angle (producing one PPI); a volume scan (see VCP) is the complete set of sweeps at every elevation angle in the pattern, producing one full 3-D snapshot of the atmosphere around the radar. *(Block 4)*

Zero isodop
The line of near-zero radial velocity in a Doppler display — everywhere the wind blows *perpendicular* to the beam, so its along-beam component vanishes. Not a region of calm: the "near-zero band curving through the storm" in Block 3's hurricane case is the zero isodop of a strong, rotating wind field. *(Block 3)*

Spectrum width
The third of the classic single-pol "base moments" alongside reflectivity and radial velocity: a measure of how much the velocities of the scatterers *within one resolution volume* spread around their mean — large in turbulence, strong shear, or mixed motion; small in smooth, uniform flow. Mentioned in Block 1's overview of what a radar measures; not otherwise used in these notebooks. *(Block 1)*

Squall line A line of thunderstorms, often fast-moving with a sharp leading edge (gust front); used in Block 4 as the scan-strategy design case, because its rapid low-level evolution rewards fast update times over deep vertical coverage. *(Block 4)*

Supercell A long-lived, rotating thunderstorm, one of the most intensely studied storm types in radar meteorology because of its association with large hail, damaging wind, and

tornadoes; used here as the real-data example storm in several widgets. *(Block 2)*

7 Master Notation Table

Symbol	Meaning	Block(s)
c	speed of light, 2.998×10^8 m/s	1–5
a_e	true Earth radius, 6371 km	1
λ, f	wavelength, frequency	1,2,3,4
D	antenna dish diameter	1,2,4
θ	half-power beamwidth (deg)	1,2,4
ϕ, elev	elevation/tilt angle (deg)	1,4
r	slant range (km)	1–5
$h(r)$	beam-center height (km)	1,4
k, R_e	effective Earth radius factor / value	1
N, dN/dh, M	refractivity (Block 1 sense), its gradient, modified refractivity	1
PBB, CBB	partial / cumulative beam blockage	1
τ	pulse length (μs)	2,3
Δr	range resolution, $c\tau/2$	2,3
P_t, G, η	transmit power, antenna gain, efficiency	2,4
Z, $Z_{\min}(r)$	reflectivity, minimum detectable reflectivity	2,4
k_B, T_0, NF	Boltzmann constant, noise temperature, noise figure	2
N	(three uses) refractivity; receiver noise power; drop number concentration	1; 2; 5
$\lvert K_w\rvert^2$	dielectric factor of water	2
V	resolution volume (km^3)	2
PRF, PRT	pulse repetition frequency / time	3
$r_{\max}$	maximum unambiguous range	3
v_{Nyq}, $v_{\max}$	Nyquist velocity	3
v_{true}, v_{disp}, v_{ext}	true / displayed / dual-PRF-extended velocity	3
a, b, $k(R)$	attenuation coefficients and specific attenuation	4
PIA	path-integrated attenuation (dB)	4
VCP, T_{vol}, rpm	volume coverage pattern, update time, rotation rate	4

Symbol	Meaning	Block(s)
H, V	horizontal / vertical polarization	5
Z_{DR}	differential reflectivity (dB)	5
ρ_{hv}	correlation coefficient	5
Φ_{DP}, K_{DP}	differential phase / its range-slope	5
b/a	raindrop axis ratio	5
R	rain rate (mm/h)	1,4,5

8 Physical Constants and Reference Values

Quantity	Value
Speed of light, c	2.998×10^8 m/s
True Earth radius, a_e	6371 km
Standard effective Earth radius factor, k	4/3 ($dN/dh \approx -39$ N/km)
Ducting onset, dN/dh	-157 N/km
Boltzmann's constant, k_B	1.380649×10^{-23} J/K
Standard noise reference temperature, T_0	290 K
Dielectric factor of water, $\lvert K_w \rvert^2$	≈ 0.93
Typical NEXRAD (S-band) parameters	$f = 2.8$ GHz, $D = 8.5$ m, $P_t = 750$ kW, $\theta \approx 0.92^\circ$
S-band wavelength	$\lambda \approx 10.7$ cm
C-band wavelength	$\lambda \approx 5.4$ cm
X-band wavelength	$\lambda \approx 3.2$ cm
Z–R relation used throughout	$Z = 200\,R^{1.6}$ (Marshall–Palmer-type)
R–K_{DP} relation used in Block 5	$R = 44\,K_{DP}^{0.82}$
Attenuation relation, S-band	$k = 0.0001\,R$ dB/km one-way
Attenuation relation, C-band	$k = 0.006\,R$ dB/km one-way
Attenuation relation, X-band	$k = 0.01\,R^{1.3}$ dB/km one-way (super-linear in R)